DIGITAL DATA & STORAGE SYSTEMS

Understanding FAT & NTFS

Microsoft File Systems — the Basics

FOUNDATIONS

Why File Systems Matter

A file system is the set of rules a disk uses to remember where every file's data physically lives — and how to find it again instantly.

FAT

Simple, table-based file system.

Older — no built-in security.

NTFS

Modern, secure file system.

Default on Windows today.

What Is FAT?

FAT (File Allocation Table) was developed by Microsoft and is one of the earliest file systems used to store and organize files on storage devices.

1977

Developed by Microsoft

Table

Maintains a File Allocation Table recording the location of every file

Wide Support

One of the most widely supported file systems

Versions of FAT

FAT12

Used in floppy disks (up to 32 MB)

FAT16

Used in early hard drives (up to 2 GB)

FAT32

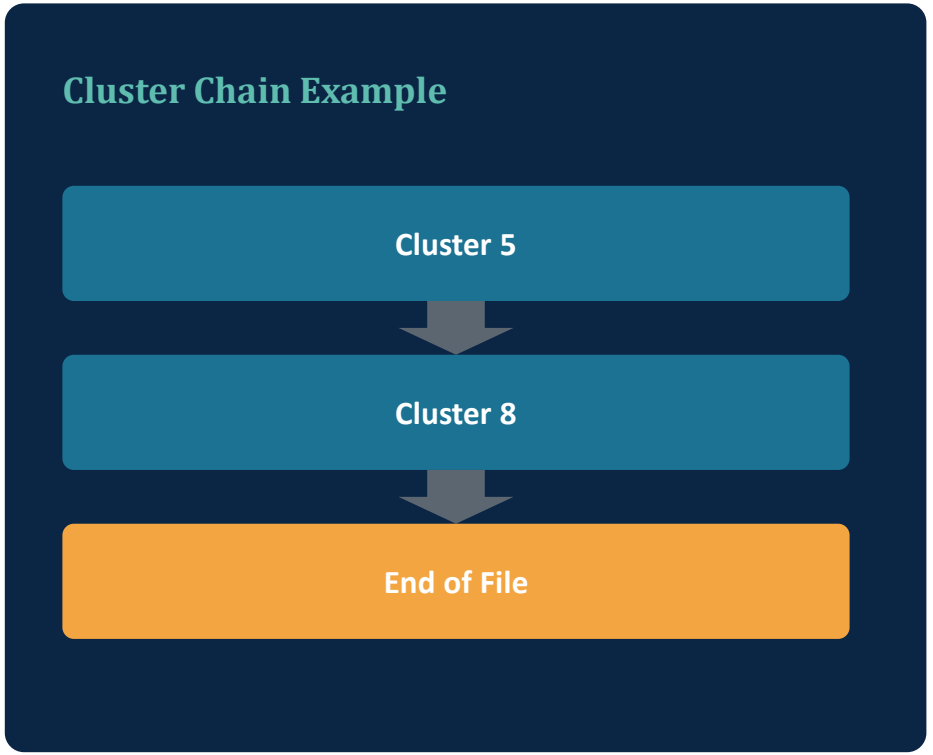
Supports larger partitions (up to 2 TB); files up to 4 GB

exFAT

Designed for flash drives and SDXC cards; supports files larger than 4 GB

How FAT Works

1. The disk is divided into clusters
2. Each file is stored in one or more clusters
3. The File Allocation Table keeps track of which clusters belong to each file
4. When a file is deleted, its clusters are marked as free for reuse



FAT — Features

- Simple structure and easy implementation
- Highly compatible with different operating systems
- Easy to format and repair
- Fast file access for small storage devices
- Requires minimal system resources

FAT — Advantages & Disadvantages

Advantages

- Supported by Windows, Linux, macOS, cameras, TVs, and gaming consoles
- Suitable for USB drives and memory cards
- Low overhead and fast performance on small disks
- Easy data recovery due to simple structure

Disadvantages

- Maximum file size limited to 4 GB (FAT32)
- No built-in security or file permissions
- No encryption or compression
- Higher risk of file corruption after improper shutdowns

FAT — Applications

- USB Flash Drives
- SD & microSD Cards
- Digital Cameras
- MP3 Players
- Embedded Systems
- Bootable USB Drives

What Is NTFS?

NTFS (New Technology File System) was introduced by Microsoft in 1993 with Windows NT, and is the default file system for modern Windows operating systems — designed for improved security, reliability, and performance.

1993

Introduced with Windows NT

Default

Used by modern Windows systems today

Secure

Built for security, reliability, performance

NTFS — Features

- Supports very large files and disk volumes
- Journaling for crash recovery
- File and folder permissions using Access Control Lists (ACLs)
- File compression
- File encryption using Encrypting File System (EFS)
- Disk quotas to limit user storage
- Supports symbolic links and hard links
- Automatic error detection and recovery

How NTFS Works

1. Stores file information in the Master File Table (MFT)
2. Uses metadata to organize files efficiently
3. Journaling records file system changes before they are written
4. Can recover quickly after unexpected power failures

NTFS — Advantages & Disadvantages

Advantages

- High reliability and fault tolerance
- Better security with user permissions
- Supports huge files and partitions
- Reduces data loss through journaling
- Better performance on modern SSDs and HDDs

Disadvantages

- More complex than FAT
- Slightly higher memory and CPU usage
- Limited write support on some non-Windows systems
- Not ideal for devices requiring universal compatibility

NTFS — Applications

- Windows Operating Systems
- Internal Hard Drives
- SSDs
- Enterprise Servers
- Large Data Storage
- Business & Corporate Computers

COMPARISON

FAT vs NTFS at a Glance

Feature	FAT	NTFS
Core structure	File Allocation Table	**Master File Table (MFT)**
Security / permissions	None	**Full ACLs**
Journaling	No	**Yes**
Max practical file size	4 GB (FAT32)	**Very large (TB+)**
Encryption	No	**Yes (EFS)**
Compression	No	**Yes**
Crash recovery	Poor	**Strong**